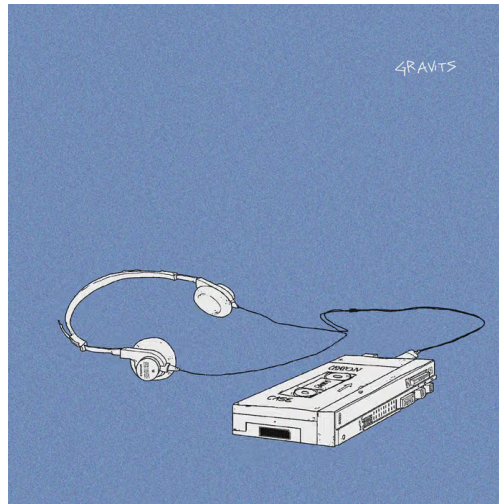


Tugas 1 IF4020 Kriptografi

Kriptanalisis pada Algoritma Kriptografi Klasik



Disusun oleh: Kelompok 3

Dita Maheswari 13523125
Naufarrel Zhafif Abhista 13523149

Program Studi Teknik Informatika
Sekolah Teknik Elektro dan Informatika
Institut Teknologi Bandung
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Pernyataan Orisinalitas

Tugas dan laporan ini disusun sepenuhnya tanpa bantuan kecerdasan buatan (*Generative AI*), melainkan hasil pemikiran dan analisis mandiri.

Dita Maheswari
13523125

Naufarrel Zhafif Abhista
13523149

Bandung, __ September 2026

Bab 1

Pendahuluan

1.1 Latar Belakang

Marisa sedang berusaha menyusun sebuah arsip rahasia yang memuat pemetaan wilayah, entitas penguasa, dan kronologi insiden penting di Gensokyo. Berkas ini merupakan gabungan dari catatan riset pribadinya serta dokumen-dokumen sensitif yang berhasil ia kumpulkan dari berbagai pihak. Demi menjaga keamanan informasi, seluruh naskah di dalam arsip tersebut dienkripsi dengan berbagai jenis *cipher* klasik. Sebagai rekan riset yang dipercaya, tugas kami adalah membantu Marisa mendekripsi serangkaian dokumen dalam arsip ini agar seluruh pengetahuan rahasia tersebut dapat terungkap.

1.2 Ringkasan Soal

Laporan ini terdiri atas lima soal wajib dan satu soal bonus. Setiap soal menuntut kriptanalisis terhadap sebuah *cipher* klasik yang berbeda.

No.	Judul	Algoritma <i>Cipher</i>
2.1	Beast Metropolis	Substitusi abjad-tunggal (<i>monoalphabetic</i>)
2.2	Horizon Chartreuse	Vigenère <i>Cipher</i>
2.3	Domain of Hopes and Dreams	Playfair <i>Cipher</i>
2.4	Trinitarian Rhapsody	Affine <i>Cipher</i>
2.5	Silent Goddess	Hill <i>Cipher</i>
3.1	Ex Machina (Bonus)	Enigma Machine M3 (termodifikasi)

Tabel 1.1: Daftar soal dan algoritma *cipher* yang digunakan.

1.3 Ketentuan Pengerjaan

Sesuai spesifikasi tugas, seluruh proses enkripsi pada *plaintext* asli dilakukan setelah teks diubah menjadi huruf kapital dan hanya pada karakter alfabetik [A-Z]; karakter lain (angka, spasi, tanda baca) dibuang dan tidak dienkripsi. Untuk soal wajib, kriptanalisis dilakukan hanya dengan *approach* yang telah diajarkan di kelas. Bantuan skrip program, lembar kerja, atau alat bantu komputasi lain diperbolehkan untuk menghitung frekuensi karakter dan pola.

Bab 2

Problem Set (Wajib)

2.1 Beast Metropolis

Deskripsi Soal

Di antara megahnya pencakar langit Alam Satwa yang dipenuhi persaingan liar, Marisa menemukan sebuah lembaran catatan yang membahas fasilitas misterius dengan struktur yang menyerupai lubang kunci. Namun, catatan mengenai benteng perlindungan tersebut kini tidak dapat dibaca karena seluruh teksnya telah dienkripsi menggunakan *cipher substitusi abjad-tunggal* (*monoalphabetic cipher*). Untuk dapat memahami struktur dari kawasan terisolasi itu, dokumen riset ini harus dikembalikan ke dalam *plaintext* seperti semula.

Bantulah Marisa mendekripsi *ciphertext* tersebut meskipun kunci pemetaannya tidak diketahui. Gunakan teknik **Analisis Frekuensi** serta analisis pola kata umum dalam **bahasa Inggris** untuk memecahkan sandi monoalfabetik ini.

Berkas lampiran:

- *Attachment:* https://drive.google.com/file/d/14DWEicqv6_2K00cA9mbXf_ZwhPrEiak/view?usp=sharing
- md5sum: dd67736bf3c6ef6db2d5f014d55d870a
- sha256sum: a6478218a83091712dc99972553f89e92da8436b2038c5b93f782268742b4972

Berkas *Ciphertext*

```
UMRQARPQNCKICIRQQIRPVQIRZABXUASQTVUXQYQIODFDQIMZIHQSUIRDUOQUACIWWQTRQAIGUOUIQJUNOXQTMFUITUCAQSCDICIVXZKQ
FDMZIDMRDFZVPCITUFZAUCIUUAUOAMQVRZAQRPFDMZIDMRTZCDRTZYBUNUCIFCYZSUWUFBDROAQMQRZAQRPFDMZIDMZGCNURQYZB
UNUCIDFUBUNUDFUBUNUOAMQVRZAQUIKRPQFDMZIDMCTPCHZRUCCIFUIKUMZFZDFUOAMQVRZAQURRPQTUNQRCNQRPAQWQAQUXTDYQIOD
FDQIMZICIVPZSDFZAQSCDIUIKFBZTPZMDAQJUNOXQRPQUIVCQIRRDNHTDMNDRTPCCIUWUCIRPQVCRBDMUTUPCFBDRDOAMQVRZAQUIKRP
QUIVCQIRRDNHTDMRDNCFUSUCIRPQVCRBDMDFUBUNUUNDISDRPQATRPQXUASQTRDIQCTRPQRDFZRDNUUIVCQIRRDNHRPQRDRUXXQISRPMR
PQNDZIKCTUHDZRRWDPZIKAKQCSPRBNQRQATWPCVPCTNDAQRPUIRPAQRCNQTXUASQARPUIRPQXUASQTRNDZIKRDNDHMDZRCSPRBNQR
QATCIRPQXURQAQACDKMRQBUBDCOQACDKUIKRTUAQUUIKVUUVCRBUAQMUSAQURQARPUIRPQXURRQAUXRPDZSPRPQNDZIKRDNHTWCR
PUMADIRUIKHUVFVCAVZXUATPUOQWQAQUXAKUBTVURRQAQKRPADZSPDZRGUOUIKZACISRPQBUBDCOQACDKRPQQAQXBMADNTMRPQYQIODF
DQIMZIWQAQNDTRXBEQARCVUXVUEQTRBXQTRDIQVPUNHQATWCRPHUNHDDTPUOQKWDDKQIVDMMCTWPCVPWQAQKCMMAQIRMADNRPNQNDZIKR
DNHTKZACISRPQBUBDCOQACDKNDADQDEQAQXABYQIODFDQIMZIPUKVDNNDIMQURZAQTTZVPUTUXDWQAUIKWCKQAMADIROUARVNDNOUAQKRDR
PQHUVFVCAVZXUAOUARUIKRPQVDNHCIURCDIUIKODTCRCDIDMSAUEQSDDKTWAQRPQTUNQCITCKQRPQVDMMCIWQAQGUQKUIKNCAADATWPCX
QDZRTCKQRPQVDMMCIWQAQXUASQIZNHQADMRAUISZXUAQSKQCECIHQUTRNCADATCADIWQUODITUSACVZXRXAUXUIMCTPCISRDDXTU
IKTDDIAQSUAKCISRPCTRUCCVPCADTPCAUCTPCHQCQEQRUPURRRPURRCNQEUAACDZTMDAVQTCIFCIDXQKHBRRPBUNURUCFCISKDNUIKRPQ
TQRDCIXUIKTQUMDANQKUIUXCUIVQUIKDHURUCIQKACADIUIKEUACDZTVZXRXAUXAQXCVTMADNFDAQUHBKMQMURCISMDAVQTTZVPUTIZFDFZ
```

UIKRPQCRDFCISKDNTWPCVPVDIRADXXQKRPQSQIFUCTQUDIRPQDRPQAPUIKBDTPCADFDIKDUXTDNQIRCDIQKRPURNDRTRMRPQYQIODFDQIM
 ZIHZCXRDI MXRUXUIKWAQWQXDA SUICYKUIKRPQNDZIKTWQAQXTDH ZCXR XUBQA HXBUBQAUMRQA VDTDXCKURCDIVDITCKQACISRPQT XD
 OQUIKUAUISQNIQIRDMRPQNDZIKTVCAVZXUANDZIKTWQAQVDTTRAZVRQKCIUKKCRCDIHADIYQNC AADATMADNNUCIXUIKVP CUIUIKRPQWUGC
 IKQIMADNRPTUNQOQACDKCIKCVURQRPURPQAQWUTUIQJVPUI SQHQRWQKIRPQWVDVZIRACQTURRPURRCNQRPQAQMDAQCRCCTOQVZXURQK
 RPURRPQQA XBUOQQAUIVQDMRPQYQIODFDQIMZINUBPUEQHQCICIMXZQIVQKHBVCECXQISCIQQACISRPQVIDXDSB MADNNUCIXUIKVP CUIU
 IKRPQODXCRCVUXUIKCKQDSCVUXCIMXZQIVQCTAQMXQVRQK CIRPQGD CIRPVNITRAZVRCDIDMRPQYQIODFDQIMZIHBEUACDZTUXXCUIVQNN
 NHQATUTOADMDMRPQUXXCUIVQURRPQTUNQR CNQRPQAQAQXUTDRDNHTMCEQMDZAU IKRPAQQDMRPQFUNQINZICIRPQMA DIRM RPQDPBZUI
 RDNH C I QUTRQAIGUOUIKZACISRPQTUNQOQACDKUTCTPCHZRUCFDMZIUNDISRPQNRDNHMC EQCTR PQQUAXCQTRFDMZICIQUTRQAIGUOUIPTCT
 RDABCIKCVURCISUODTTCHXQVDIIQVRCDIHQRWQKIRPQFUYZTUOADECIVUIKRPQBUNURUCFCISKDNURRPURRCNQ CIRPQMDZARPVQIRZABP
 DFZACFZAQSCDIFUIRDAQSCDIRDUFUCAQSCDIUIKRPDPFZAQSCDIWQAQNUCIXBVDNODTQKDMMADIRUIKAQUARDNHTWPCVPSAUKZUXXBH QVUN
 QYQIODFDQIMZIRWDRDRAQQSQIAURCDITXURQAASUA K CISRPCTBUNURDRUFQAZODCIRKQDZRRPURUXRPDZSPRPQDFGFCUIKICPDITPD
 FCNQIRCDIQKRPURRPQMDZASQIAUXTU IKGUOUIQTQRUFQYZIUIKDRPQATPUNQASQKQUTRGUOUICIRDRPQRQAACRDA BDMRPQGUOUIQTQND
 IUA V PBRPADZSPNZXRCOXQQJQKCRCDITPQHQCQEQRKRPURCWRUTUVVRZUXBFZIZIDFZICXDVRQK CIRPQIDHCOXUCIRPURMDANQKUIUXXC
 UIVQCIQUTRGUOUI NUCIXBVDITCTRCISDMAQUASAEQTCIRPQOUTRUIKRPURUMRQARPQKQURPDMPCNCFDRPQBXDTRRPQHURRXQWCRPRPQBU
 NURUCFCISKDNDAIQSDRCURQKOQVQZIKARPPQXQKATPCODMRPQBUNURDFCISTPCOCIRPQKIRPQUXXCUIVQDMRPQK DSTXUEQFCISKDNU
 IKRPQUXXCUIVQDMRPQECXPDATQFCISKDNQASQK CIRDDIQMDANCISRPQGUOUIQTQNDIUA V PBRPQIRPQXQKQATDMQUTRUIKWTRGUOUIW
 QAQKCTRCISZCTPQKHBRPQYQIODFDQIMZUIKRPQMA DIRUIKAQUARDNHTMADNRPQQIKDMRPQMDZARPVQIRZABDRPQCMRPVQIRZABXUASQ
 YQIODFDQIMZIAQTQNHXCISDFMZICINDZIRRUCKUHQSUIRDUOQUACIOAQTQIRKUBSZINUOAQMQVRZAQXUASQYQIODFDQIMZIPUEQXUTDHQ
 QIHZCXR CIOXUVQTTZVPUBUNUIUTPCOAQMQVRZAQCHUAUF COAQMQRZAQIUIKVPCHUOAQMQVRZAQUMRQARPQNCMDZARPVQIRZABRPQKUWU
 RDNHTDACSIXXBVQIRQAQKUADZIKRPQBUNURDFDMZICIRPQTDZRPQUTRDMRPQIUUAHU TC IUIKRPQABZHQIFDMZISAUKZUXXBTPCMRQKRD
 RPQKUCTQIFDMZICIRPQIDARPRPQXUASQYQIODFDQIMZIDMKUWUWUTRPQMZA ZVPCFDMZIVXZTRQAUIKNDY ZRDNHTRPQVQIRQADMSAUECRB
 DMRPQSAQURFCISTRDNHTPCMRQKUSUCICIRPQCMRPVQIRZABDRPQIUVVCIRVCRBRDNH SADZOUIKRPQPZIKAKRDISZQKHA KRDNHSADZO
 CIRPQDTUFUOXUCIAQSUAKCISRPQAQUTDITMDARPQNZXRCOXQRAUITM QATMRPQTQXUASQYQIODFDQIMZIHUTQKDIAQTQUAVPDIRPQFZTPC
 GCUIKICPDITPZGCRPQGUOUIQTQVUUKQNCVVDNNZICRBPUTOADODTQKECQWTTZVPUTRPQRPDABDMKBIUTRBUXRAIURCDIUIKRPQRPDAB
 DMQLZQTRAUIRACHQTV DILZQACISKBIUTRCQTRDQJOXUCIDIRPQDRPQAPUIKPCADTQFUYZDOADODTQKRPURRPQTTRAZVRZAQTDMRPQTUTUFC
 FDGCTPCYZFCUIKNUF CIDIUVCQIRRDNHSADZOTWQAQTCNCXUAUIK MADNR PQXURQMDZARPVQIRZABDRPQTQVDIKPUXMDMRPQCMRPVQIRZA
 BRPQPQKUTDMRPQMDZAU IVCQIRRDNHSADZOTGDCIRXBOXUBQKUADXQCIRPQBUNURDFCISTPCOUMRAKQURPRPQBKQCMCQKUTRPQSUAUCUI
 KQCRBDMRPQBUNURDADBUXODWQAUIKUXTD OADODTQKRPURRPQVDITRAZVRCDIDMRPQXUASQYQIODFDQIMZIPUKUKRQAAQIRQMMQVRDIEUA
 CDZTAQSCDITURRPQTUNQR CNQ CIRPQTQVDIKPUXMDMRPQCMRPVQIRZABIDXUASQTVUXQYQIODFDQIMZIWQAQVDITRAZVRQKCIUAQUTT ZVP
 UTFCHCOADECIVQZOOQANUBBQUIKPBZSUOADECIVUIKRPQT VUXQDMRPQYQIODFDQIMZICISDFCTPCVPCKDSAU KZUXXBKQVAQTQKUVVDAK
 CISRDRPQCITVACORCDITDIGCGCRPQCADITWDAKZIQUAR PQK MADNR PQCIUWUNDZIRUCIFDMZUIKRPQXUASQTDWDAKZIQUAR PQK MADNR PQQR
 UMZIU BUNUFDMZIQNOQADABZABUFZURRPURRCNQAQSUAKQKPCNTQXMTRPQFCISDMRPQWDAXKUIKPCTUZRPD ACBWTATAQMXQVRQK CIRPQMU
 VRRPURDIXBPQVDIRCIZQRDHZCXKXUASQYQIODFDQIMZICIRPQGUOUIQTQUAVPCOQXUSD CIRPQTCJRPVQIRZABRPQPDACYDIRUXVUEQTRB
 XQTRDIQVPUNHQADMRPQFCIUCTBTQRNHQSUIRDHQDOZUAC YQKCIYQIODFDQIMZIU RRPQTUNQR CNQRPQBHSUIRDUOQUACIUAQUTT ZVP
 TTQRTZOADECIVQDWUACOADECIVQZGCOADECIVUIKQE QICIFUIRDAQSCDIWPCVPWQAQVDITCKQAQKUTHUTQTMDARPQQTURHXT PNQIRDM
 RPQNOQADAFQCRUCUNDISRPQNZQIDPUKUOUARCVZXUAXBXUASQIZNHQADMYQIODFDQIMZIVDNOUAQKRD DIXBRPCARBIQICIFDYZFQOADE
 CIVQKZACISRPQTUNQOQACDKZQIDUXDIQPUKICIQRBTEQIWPXCQIFUIRDRPQAQWQAQRWDPZIKAKT CJRQJUTHDRPFCIDUIKDCYPUISKQ
 VAQUTQKDIXBRPQFUIRDAQSCDIDIRCIZQRDVIDITRAZVRXUASQYQIODFDQIMZIRPADZSPDZRRPQTCJRPVQIRZABCIKCVURCISRPURRPQBU
 NURDNDIUA V PBRURRPURRCNQAQXCQKQKUECXBDIRPQUTRQAIFCISKDN CIHDPQVDIDNCVUIKNCXCRUABUTOQVRTMADNR PQKIDMRPQTCJRP
 VQIRZABDRPQHQS CII C ISDMRPQTQE QIRPVQIRZABXUASQTLZUAQIUKVCAVZXUARDNHTAQOXUVQKRPQMA DIRUIKHQVUNQRPNUCITRAQUNC
 IRPQNCKKXQDMRPQTQE QIRPVQIRZABRPQBQEDXEQK CIRDDVRUSDIXRDNHTAQSUAKCISRPQAQUTDIMDARPQVQTTURCIDMVDITRAZVRCDID
 MYQIODFDQIMZIRUCCVPCADTPCAUCTPCHQCQE QTRPURCRCTAQXURQKDRPQAQMDANTDMVADWIOACIVQTPDRDFZUIKTD SUIDZNUF DUTWQXX
 UTRPQQTRUHXCTPNQIRDMRPQIURCDIUXNUIZMUVRZACISTBTRQNWPCXQRPQMCIXUYQIODFDQIMZIU AQHQCQEQRD HQUTUNUNDZIRUCIFDM
 ZIUIKADISNUVPCVXCNTRPURRPQFDMZIWQAQHZCXRCIRPQMCATRPXUMMRPQTQE QIRPVQIRZAB

Langkah Penyelesaian

Monalphabetic cipher rentan terhadap analisis frekuensi. Jika kriptanalisis mendapatkan frekuensi yang mendekati distribusi huruf pada kata-kata di Bahasa Inggris, sebuah *ciphertext* dapat diterka dengan mencocokkan monogram, bigram, bahkan hingga trigram.

Langkah-langkah yang dilakukan:

1. Lakukan penghitungan frekuensi monogram pada *ciphertext*.

```

uv run 21/solver.py
rank cipher cipher% -> plain plain%
1 Q 12.31 -> E 12.02
2 R 9.27 -> T 9.10
3 U 8.83 -> A 8.12
4 I 8.60 -> O 7.68
5 D 8.06 -> I 7.31
6 C 6.64 -> N 6.95
7 A 6.10 -> S 6.28
8 P 5.61 -> R 6.02
9 T 4.30 -> H 5.92
10 K 3.56 -> D 4.32
11 Z 3.56 -> L 3.98
12 M 3.07 -> U 2.88
13 V 2.89 -> C 2.71
14 X 2.81 -> M 2.61
15 N 2.73 -> F 2.30
16 S 2.01 -> Y 2.11
17 O 1.96 -> W 2.09
18 B 1.78 -> G 2.03
19 F 1.77 -> P 1.82
20 H 1.34 -> B 1.49
21 W 0.96 -> V 1.11
22 E 0.64 -> K 0.69
23 Y 0.59 -> X 0.17
24 G 0.39 -> Q 0.11
25 J 0.15 -> J 0.10
26 L 0.05 -> Z 0.07

```

Kita dapatkan pemetaan (kasar) berikut:

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	SGNIKPQBOJZUFWRETYHACVMXL

2. Untuk memperkuat dasar pencocokan, hitung juga frekuensi bigram dan trigram pada *ciphertext* (`freq_ngram(ciphertext, n)` dengan $n = 1, 2, 3$), lalu bandingkan sepuluh besarnya dengan referensi Bahasa Inggris.

Monogram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	Q	12.31	E	12.02
2	R	9.27	T	9.10
3	U	8.83	A	8.12
4	I	8.60	O	7.68
5	D	8.06	I	7.31
6	C	6.64	N	6.95
7	A	6.10	S	6.28
8	P	5.61	R	6.02
9	T	4.30	H	5.92
10	K	3.56	D	4.32

Bigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RP	3.88	TH	3.56
2	PQ	2.99	HE	3.07
3	CI	2.21	IN	2.43
4	UI	1.83	ER	2.05
5	QI	1.80	AN	1.99
6	AQ	1.70	RE	1.85
7	IR	1.64	ON	1.76
8	QA	1.49	AT	1.49
9	DI	1.42	EN	1.45
10	IK	1.39	ND	1.35

Trigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RPQ	2.91	THE	1.81
2	UIK	1.03	AND	0.73
3	QIR	0.62	ING	0.72
4	MZI	0.62	ENT	0.42
5	DMR	0.61	ION	0.42
6	MRP	0.61	HER	0.36
7	IRP	0.59	FOR	0.34
8	CIS	0.56	THA	0.33
9	QAQ	0.52	NTH	0.33
10	KRP	0.51	INT	0.32

Bentuk distribusinya memang tidak identik urutan-per-urutan, tetapi tiga huruf/pasangan/tripel tersering pada *ciphertext* (Q, R, U dan RP, PQ dan RPQ) menempati rentang persentase yang sebanding dengan huruf/pasangan/tripel tersering Bahasa Inggris (E, T, A dan TH, HE dan THE). Hal ini konsisten dengan hipotesis awal bahwa *cipher*-nya adalah substitusi abjad-tunggal murni terhadap teks Bahasa Inggris, sehingga pencocokan berbasis frekuensi dapat dilanjutkan.

3. Lakukan substitusi berdasarkan pemetaan (kasar) yang telah dibuat.

AUTESTREFNDONOEETOTRCEOTLSGMASYEHKAMEXEOWIPIEOULOBEOAOTIAWWEASNOVEHTESQAWAOEJAFWMEHIUPAOHANSEYNIONO
CMLDEPIULOIUTIPLCRNNOHAPLSANOASAWSEUECTLSETREPIULOIUTHNLTHLXLGAFANOPNXYAVAPGITIWSEUECTLSETREPIULOIUI
LQNFATEXLGAFANOIPAGAFAPAGAFWSEUECTLSEAODTREPIULOIUNHRLTANNOPAOAULPLIPAWSEUECTLSEATTREHAFETNFETRE
SEVESEAMHIXEOWIPIEOULONOCRLIPLSEYNIOAODPGLHRLUISEJAFWMETREAOACNEOTTIFBHIUFIITHRNOAVANOTRECNTGUAHARN
PGITIWSEUECTLSEAODTREAOACNEOTTIFBHIUTIFNPAYANOTRECNTGIUPAGAFAAFIOTITRESHTREMASYEHTIOENHTRETITPLTIFAAOC
NEOTTIFBTRETITAMMEOYTRIUTREFILODNHABILTTVIRLODSEDENYRTGFETESHVRNCRNHFISETRAOTRSEETNFEHMASYESTRAOTREMA
SYEHTFILODTIFBIUABILTENYRTGFETESHNOTREMATESWESNIDIUTREGAGINWESNIDAODNTHASEAAODCAWACNTGASEUASYSEATESTR
AOTREMATTESAMTRILYRTREFILODTIFBHVNTRAUSIOTAODBACPCNSCLMASHRAWVESEAMSEADGHCAETTESDTRSILYRILTQAWAODLSN
OYTREGAGINWESNIDTREEASMGUISFHIUTREXEOWIPIEOULOVESEFIHTMGKESTNCAMCAKEHTGMEHTIOECRAFVBESHVNRBAFBIHRAW
DVIIDEOCIUUNOHVRNCRVESEDNUVESEOTUSIFTREFILODTIFBHDLSNOYTREGAGINWESNIDFISEIKESEASMGXEOWIPIEOULORADCIFF
IOUEATLSEHHL CRAHAMIVESAODVNDUSIOTWASTCWFASEDTITREBACPCNSCLMASWASTAODTRECIFBNOATNIOADWIHNTNIOIUYSA
KEYIIDHVESETREHAFENOHNDRECIUUNOVESEQADEAODFNSSISHVRNMEILTHNDETRECIUUNOVESEAMASYEOLFBESIUTSNAOYLMASE
DYEDKNNOEBEAHTFNSSISHNSIOVEAWIOHAYSNCMLTMSAMAODUNHRNOYTIIMHAODHIIIOSEYASDNOYTRNHTANNCRNSIHRNSANHRNBEMN
EKEDTRATATTRATTFEKASNILHUISCEHNOPNOIMEDBGTREGAFATANPNYDIFAODTREHETINOMADHEAUISEFEDAOAMNNOCEAODIBTA
NOEDNSIOADKASNILHCLMTLSAMSEMNCUSIFPISEABGDEUEATNOYUISCEHHL CRAHOLPIPLAODTRENTIPNOYDIFHVRNCRCIOTSIMME
DTREYEOPANHEATOTREITRESAODGIHRNSIPIODIAMHIFEOTNIOEDTRATFIHTIUTREXEOWIPIEOULOBMLNMTIOMATMAODVESEVEMMI
SYAONXEDAODTREFILODHVESEAMHIBLNM TMAGESBGMAGESAUTESCIOHIMNDATNIOCIHNDENSOYTREHMIWEAODASSAOYEFEOIUTRE
FILODH CNSCLMASFILODHVESECIOHTSLCTEDNOADDNTNIOBSIOXEFNSSISHUSIFFANOMAODCRNOAAODTREVAQNODEOUSIFTREHAFEW
ESNIDNODNCATETRATTRESEVAHAOEJCRAOYEBETVEEOTRETVICILOTSNEHATTRATTNFETRESEUISENTNHHWECLMATEDTRATTREEASM
GAWWEASAOCEIUTREXEOWIPIEOULOFAGRAKEBEEONUOMLEOCEDBGCNKNMEOYNOEESNOYTECROIMIYGUSIFFANOMAODCRNOAAODTREW
IMNTNCAMAODNDEIMIYNCAMNOUMLEOCENHSEUMECTEDNOTREQINOTCIOHTSLCTNIOIUTREXEOWIPIEOULOBGKASNILHAMNNOCEFEF
BESHAWHSIIUIUTREAMNNOCEATTREHAFETNFETRESEASEAMHITIFBHNKEUILSAODTRSEEIUTREPAFEOLFONOTREUSIOTIUTRERIL

GLAOTIFBNOEAHTESQAWAODLSNOYTREHAFEWESNIDAHNRNBLTANPIULOAFIOYTREFTIFBUNKENHTREEASMNEHTPIULONOEAHTESO
 QAWAOHRNHTISGNODNCATNOYAWIHHNBMECIOECTNIOBETVEOTREPAXLHAWSIKNOCEAODTREGAFATANPNODYIFATTRATTNFENOTRE
 UILSTRCEOTLSGRIPLSNPLSEYNIOPAOTISEYNIOTIPANSEYNI OAODTIRIPLSEYNI OVESEFANOMGCIFWIHEDIUUSIOTAODSEASTIFBH
 VRNCRYSADLAMMBECAFEXEOWIPIEOULOTVITITRSEEYEOESATNIOHMATESSEYASDNOYTRNHGAFATITAPESLWINOTEDILTTRATAMTR
 ILYRTREPIQNPNAODONRIOHRIPNFEOTNIOEDTRATTREUILSYEOESAMHAODQAWAOEHETAPEXLOAODITRESHRADFESYEDEAHTQAWAONO
 TITRETESNTISGIUTREQAWAOEHEFIOASCRGTRSILYRFLMTNWMEJWEDNTNIOHREBEMNEKEDTRATNTVAHACTLAMMGLOLOIPLONMIC
 ATEDNOTREOIBNWMANOTRATUISFEDAOAMMNAOCENOEAHTQAWAOFANOMGCIOHNHTNOYIUSEASYSAKEHNOTREWAHTAODTRATAUTESTRE
 DEATRIURNFPITREGMIHTTREBATTMEVNTRTREGAFATANPNODYIFISOEYITNATEDWEACELODESTREMEADESHRNWIUTREGAFATIPNOY
 HRNWNOTREODTREAMMNAOCEIUTREDIYHMAKEPNOYDIFAODTREAMMNAOCEIUTREKNMRISHEPNOYDIFESYEDNOTIOEUISFNOYTRE
 QAWAOEHEFIOASCRGTREOTREMEADESHIUEAHTAODVEHTQAWAOVESEDNHTNOYLNHREDBGTREXEOWPIEOULO AODTREUSIOTAODSEAST
 IFBHUSIFTREEODIUTREUILSTRCEOTLSGTITREUNTRCEOTLSGMASYEXEOWIPIEOULOSEHEFBMNOYPIULONOFILOTTANDABEYAOITIA
 WWEASNOWSEHEOTDAGYLOFAWSEUECTLSEMASYEXEOWIPIEOULORAKEAMHIBEEOBLNMTNOWMACEHLCRAHGAF AOAHRNWSEUECTLSENB
 ASAPNWSEUECTLSEAODCRNBWSEUECTLSEAUTESTREFNDUILSTRCEOTLSGTREDAVATIFBHSINYNAMMGCEOTESEDASILODTREGAFAT
 IPULONOTREHILTREAHITUTREOASABAHNOAODTRESGLBEOPULYOYADLAMMGHRNUTEDTITREDANHEOPIULONOTREOISTRTREMASYE
 XEOWIPIEOULOIUDAVAVAHTREULSLNCRNPULOCMLHTESAODFXILTFIBHTRECEOTESIUYSAKNTGIUTREYSEATPNODYTIFBHRNUTEDA
 YANONOTREUNTRCEOTLSGTITREAOCEOTCNTGTIFBYSILWAODTRERLODSEDTIOYLEDNBSDTIFBYSILWNOTREIHAPAWMANOSEYASDN
 OYTRESEAHIOHUISTREFLMTNWMETSAOHUESHIUTREHEMASYEXEOWIPIEOULOBABEDIOSEHEASCRITOTREPLHRNQNAODONRIOHRLQNTR
 EQAWAOEHEACADEFNCCIFFLONTGRAHWSIWIHEDKNEVHHL CRAHTRETREISGIUDGOAHTGAMTESOATNIOAODTRETREISGIEZLEHTSAOT
 SNBEHCIOZLESNOYDGOAHTNEHTIEJWMANOIOTREITRESRAODRNSIHEPAXLWSIWIHEDTRATTREHTSLCTLSEHIUTREHAHAPNP IQHNRN
 XLPAODFAPNOIAOCNEOTTIFBYSILWHVESEHNFNMAS AODUSIFTREMA TEUILSTRCEOTLSGTITREHECIODRAMUIUTREUNTRCEOTLSGT
 REREAHIUTREUILSAOCNEOTTIFBYSILWHQINOTMGWMAGEDASIMENOTREGAFATIPNOYHRNWAUTESDEATRREGDENUNEDAHTREYLASD
 NAODENTGIUTREGAFATISIGAMWIVESAODAMHIWSIWIHEDTRATTRECIOTSLCTNIOIUTREMASYEXEOWIPIEOULORADADETESSEOTEUU
 ECTIOKASNILHSEYNIOHATTREHAFETNFENOTREHECIODRAMUIUTREUNTRCEOTLSGOIMASYEH CAMEXEOWPIEOULOVESECIOTSLCT
 EDNOASEAHHL CRAHPNBNWSIKNOCELWESFAIGEAO DRGLYAWSIKNOCEAODTREHCAMEIUTREXEOWPIEOULONOPYNHRNCRNDIYSADLA
 MMGDECSEAHEDACCISDNOYTTIRENOHCSNWTNIOHQNTRENSIOHVISDLOEASTREDUSIFTRENOAVAFILOTANOPIULO AODTREMASYE
 HVISDLOEASTREDUSIFTREETAULOAGAFAPIULOEFWESISGLSGAPLATTRATTNFSEYASDEDRNFHEMUAHTREPNOYIUTREVISMADOARNH
 ALTRISNTGVAHSEUMECTEDNOTREUACTTRATIOMGRECIOTNOLEDTIBLNDMASYEXEOWIPIEOULONOTREQAWAOEHEASCRNWEMAYINOTR
 EHNJTRCEOTLSGTRERISNXIOTAMCAKEHTGMEHTIOECRAFBSIUTREPNOANHGHTEFBYAOITIBEWIWLMSXEDNOXEOWPIEOULOATTR
 EHAFETNFETREGBEYAOITIAWWEASNOASEAHHL CRAHHTHLSIKNOCEIVASNWSIKNOCELQNWSIKNOCEAO DEKEONOPAOITISEYNIORVNC
 RVESECI OHNDESEDAHBAHEHUISTREEHTABMNRHFEOTIUTREEFWESIS PENTANAFIOYTREFLEOIRADAWASTNCLMASMGMASYEO LFBESI U
 XEOWIPIEOULOCIFWASEDTI IOMGRNSTGONOENOPIXLPEWSIKNOCEDLSNOYTREHAFEWESNIDLEOIAMIOERADONOTGHEKEOVRNMENO
 PAOTITRESEVESETVIRLODSEDH NJTEOAHBITRPNIOAODINXRAOYDECSEAHED IOMGTREPAOTISEYNIOCIOTNOLEDTICIOHTSLCTMAS
 YEXEOWIPIEOULOTRSILYRILTTREHNJTRCEOTLSGNODNCATNOYTRATTREGAFATIFIOASCRGATTRATTNFSEMNEDREAKNMGIOTREAH
 TESOPNOYDIFNOBITRECIOIFNCAODFMNTASGAHWEC THUSIFTREEODIUTREHNJTRCEOTLSGTITREBEYNOONNOYIUTREHEKEOTRCEOTL
 SGMASYEHZLASEAODCNSCLMASTIFBHSEWMACEDTREUSIOTAODBECAFETREFANOHTSEAFNOTREFNDMEIUTREHEKEOTRCEOTLSGTREG
 EKIMKEDNOTI ICTAYIOAMTIFBHSEYASDNOYTRESEAHIOUISTRECEHATNIOIUCIOHTSLCTNIOIUXEOWPIEOULOTANNCRNSIHRNSAN
 HRNBEMNEKEHTRATNTNHSEMATEDTITRESEUISFHIUCSIVOWSNOCEHRITIPLAODHIYA OILFAPIAHVEMMAHTREEHTABMNRHFEOTIUTRE
 OATNIOAMFAOLUACTLSNOYHGHTFVRNMETREUNOAMXEOWIPIEOULOASEBEMNEKEDTIBEAHAFAFILOTANOPIULO AODSIOYFACRNCMAN
 FHTRATTREPIULOVESEBLNMTNOTREUNSHTRAMUIUTREHEKEOTRCEOTLSG

4. Cocokkan melalui informasi kecocokan bigram dan trigram, lalu lanjutkan dengan iterasi yang sama: setiap kali sebuah bigram/trigram/kata umum Bahasa Inggris mulai terlihat pada hasil dekripsi semantara, huruf *cipher* yang bersangkutan diganti agar sesuai. Tiga huruf – R, Q, U – sudah benar sejak tebakan awal (langkah 1) dan tidak pernah diubah lagi. Untuk sisanya, dapat dilihat pada tabel di bawah yang disusun berdasarkan *log* dari *src/21/log/plain/* secara kronologis.

Ganti	Konteks (sebelum → sesudah)	Petunjuk
P→H	AUTESTREFNDONOETEEOTR → AUTESTHEFNDONOETEEOTH	delapan karakter pertama UMRQAR PQ → AUTESTHE, trigram THE
I→N	UTESTHEFNDONOETEEOTHCEOTL UTESTHEFNDNNNTEENTHCENTL →	mengarah ke NINETEENTH
C→I	AUTESTHEFNDNNNTEENTHCEN AUTESTHEFIDNINETEENTHCEN →	kata NINETEENTH terbentuk
A→R	AUTESTHEFIDNINETEEN → AUTERTHEFIDNINETEEN	bigram ER
Z→U	TEENTHCENTLSGMASYEHCAMEXE TEENTHCENTURYMARYEHCAMEXE →	mengarah ke CENTURY
B→Y	ENTHCENTLSGMASYEHCAMEXENW ENTHCENTURYMARYEHCAMEXENW →	kata CENTURY lengkap
M→F	AUTERTHEFIDNINET → AFTERTHEFIDNINET	kata umum AFTER THE
N→M	AFTERTHEFIDNINETEENTHCE AFTERTHEMIDNINETEENTHCE →	kata MID (AFTER THE MID-...)
D→O	EHCAMEXENPIPIENFUNBEYANTI EHCAMEXENPOPOENFUNBEYANTO →	mengarah ke ZENPOKOENFUN
T→S	NTURYMARSEHCAMEXENPOPOENF NTURYMARYESCAMEXENPOPOENF →	bigram ES
F→K	SCAPEXENPOPOENFUNBEYANTOA SCAPEXENPOKOENFUNBEYANTOA →	mengarah ke ZENPOKOEN
S→G	CENTURYPARYESCAPEXENPOKOE CENTURYPARGESCAPEXENPOKOE →	kata LARGE
O→P	GESCAPEXENLOKOENFUNBEGANT GESCAPEXENPOKOENFUNBEGANT →	kata ZENPOKOEN
X→L	NTHCENTURYPARGESCAPEXENPO NTHCENTURYLARGESCALEXENPO →	kata LARGE SCALE
W→W	TOAPPEARINGESTERNQAPANEJA TOAPPEARINWESTERNJAPANEXA →	kata IN WESTERN
G→J	RINGESTERNQAPANEJAMPLESOF RINWESTERNJAPANEXAMPLESOF →	kata JAPAN
J→X	TERNQAPANEJAMPLESOFKANSAI TERNJAPANEXAMPLESOFKANSAI →	kata EXAMPLES
Y→Z	LARGESCALEXENPOKOENFUNBEG LARGESCALEZENPOKOENFUNBEG →	istilah ZENPOKOENFUN terbentuk
E→V	WEREMOSTLYKERTICALCAKESTY WEREMOSTLYVERTICALCAVESTY →	kata VERTICAL CAVE
L→Q	LARGESZUAREANDCIR → LARGESQUAREANDCIR	kata SQUARE

Artefak

Skrip *solver* yang digunakan untuk menghitung frekuensi monogram/bigram/trigram, menyusun tebakan awal, menerapkan pemetaan hasil analisis, serta mencatat setiap hasil dekripsi ke log/plain/ dan metadatanya (pemetaan & frekuensi n-gram) ke log/metadata/. Modul ini bergantung pada `src/frequencies.py` yang berisi tabel referensi frekuensi Bahasa Inggris (dipakai bersama untuk seluruh soal substitusi/Vigenère pada laporan ini, sehingga isinya tidak diduplikasi di sini).

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List
6
7 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
8 from frequencies import MONOGRAM, MONOGRAM_ORDER
9
10 CORRECTIONS = [
11     ("R", "T"), ("P", "H"), ("Q", "E"), ("I", "N"), ("C", "I"),
12     ("U", "A"), ("Z", "U"), ("A", "R"), ("B", "Y"), ("M", "F"),
13     ("O", "P"), ("W", "W"), ("N", "M"), ("D", "O"), ("T", "S"),
14     ("X", "L"), ("G", "J"), ("F", "K"), ("S", "G"), ("J", "X"),
15     ("Y", "Z"), ("E", "V"), ("L", "Q"),
16 ]
17

```

```

18 class Solver:
19     def count(self, string: str) -> Dict:
20         data = {}
21         for c in string:
22             data[c] = data.get(c, 0) + 1
23         return dict(sorted(data.items(), key=lambda item: item[1], reverse=True))
24
25     def sort(self, string: str) -> List:
26         return list(self.count(string).keys())
27
28     def freq_ngram(self, string: str, n: int = 1) -> Dict:
29         freq_data = {}
30         total_ngrams = len(string) - n + 1
31         for i in range(total_ngrams):
32             ngram = string[i:i+n]
33             freq_data[ngram] = (freq_data.get(ngram, 0) + 1)
34         for ngram in freq_data:
35             freq_data[ngram] = (freq_data[ngram] / total_ngrams * 100)
36         return dict(sorted(freq_data.items(), key=lambda item: item[1], reverse=
37 True))
38
39     def solve(self, string: str) -> Dict:
40         cipher_order = [c for c in self.sort(string) if c.isalpha()]
41         mapping = {}
42         for cipher_letter, plain_letter in zip(cipher_order, MONOGRAM_ORDER):
43             mapping[cipher_letter] = plain_letter
44         return mapping
45
46     def replace_char(self, mapping: Dict, char0: str, char1: str) -> Dict:
47         mapping[char0] = char1
48         return mapping
49
50     def decrypt(self, string: str, mapping: Dict) -> str:
51         return "".join(mapping.get(c, c.lower()) for c in string)
52
53     def table(self, string: str) -> str:
54         cipher_freq = self.freq_ngram(string)
55         rows = ["rank cipher cipher% -> plain plain%"]
56         for i, cipher_letter in enumerate(cipher_freq):
57             if not cipher_letter.isalpha():
58                 continue
59             plain_letter = MONOGRAM_ORDER[i] if i < len(MONOGRAM_ORDER) else "-"
60             plain_pct = MONOGRAM.get(plain_letter, 0.0)
61             rows.append(
62                 f"{i+1:>4} {cipher_letter:^6} {cipher_freq[cipher_letter]:>6.2f}
63                 -> "
64                 f"{plain_letter:^5} {plain_pct:>5.2f}"
65             )
66         return "\n".join(rows)
67
68     def load_ciphertext(self, base_dir: str) -> str:
69         with open(os.path.join(base_dir, "s.txt")) as f:
70             return f.read().strip()
71
72     def write_plain(self, base_dir: str, timestamp: str, decrypted: str) -> str:
73         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
74         os.makedirs(os.path.dirname(path), exist_ok=True)
75         with open(path, "w") as f:
76             f.write(decrypted)

```

```

75     return path
76
77 def write_metadata(base_dir: str, timestamp: str, mapping: Dict, freq: Dict) -> str
78 :
79     path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
80     os.makedirs(os.path.dirname(path), exist_ok=True)
81     payload = {
82         "timestamp": datetime.now().isoformat(timespec="seconds"),
83         "mapping": mapping,
84         "freq_ngram": {str(n): freq[n] for n in freq},
85     }
86     with open(path, "w") as f:
87         json.dump(payload, f, indent=2)
88     return path
89
90 if __name__ == "__main__":
91     base_dir = os.path.dirname(os.path.abspath(__file__))
92     content = load_ciphertext(base_dir)
93
94     solver = Solver()
95     mapping = solver.solve(content)
96     for cipher_letter, plain_letter in CORRECTIONS:
97         mapping = solver.replace_char(mapping, cipher_letter, plain_letter)
98
99     decrypted = solver.decrypt(content, mapping)
100     freq = {n: solver.freq_ngram(content, n) for n in (1, 2, 3)}
101
102     timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
103     plain_path = write_plain(base_dir, timestamp, decrypted)
104     metadata_path = write_metadata(base_dir, timestamp, mapping, freq)
105
106     print("mapping (cipher -> plain):")
107     print("".join(sorted(mapping)))
108     print("".join(mapping[c] for c in sorted(mapping)))
109     print()
110     print("hasil dekripsi:")
111     print(decrypted)
112     print()
113     print(f"[tersimpan ke {os.path.relpath(plain_path)} dan {os.path.relpath(
114         metadata_path)}]")

```

src/21/solver.py

Plaintext Hasil Dekripsi

AFTERTHEMIDNINETEENTHCENTURYLARGESCALEZENPOKOENFUNBEGANTOAPPEARINWESTERNJAPANEXAMPLESOFKANSAIREGIONINCLUDE KOFUNOFTOKUCHIINSAKURAINARAPREFECTURETHEKOFUNOFTSUIOTSUZUYAMAINKIZUGAWAKYOTOPREFECTURETHEKOFUNOFUJIMATEZUY AMAINOKAYAMAOKAYAMAPREFECTUREANDTHEKOFUNOFISHIBUTAIINKANDAFUKUOKAPREFECTUREATTHESAMETIMETHEREWEREALSOZENPO KOENFUNINCHUGOKUREGIONANDKYUSHUFOREXAMPLETHEANCIENTTOMBSOFMOTOSHIINAWAINTHECITYOFASAHIKYOTOPREFECTUREANDTH EANCIENTTOMBSOFTOMIKAGAINTHECITYOFKAYAMAAMONGOTHERSTHELARGESTONEISTHETOKUTOMAANCIENTTOMBTHETOTALLENGTHOFT HEMOUNDISABOUTTWOHUNDREDEIGHTYMETERSWHICHISMORETHANTHREETIMESLARGERTHANTHELARGESTMOUNDTOBFOFABOUTEIGHTYMET ERSINTHELATERPERIODOFTHEYAYOIPERIODANDITSAREAANDCAPACITYAREFARGREATERTHANTHELATTERALTHOUGHTHEMOUNDTOMBSWIT HAFRONTANDBACKCIRCULARSHAPEWEREALREADYSCATTEREDTHROUGHOUTJAPANDURINGTHEYAYOIPERIODTHEEARLYFORMSOFTHEZENPOK OENFUNWEREMOSTLYVERTICALCAVESTYLESTONECHAMBERSWITHBAMBOOSHAPEDWOODENCOFFINSWHICHWEREDIFFERENTFROMTHEMOUND TOMBSDURINGTHEYAYOIPERIODMOREOVEREARLYZENPOKOENFUNHADCOMMONFEATURESSUCHASALOWERANDWIDERFRONTPARTCOMPARED TOT HEBACKCIRCULARPARTANDTHECOMBINATIONANDPOSITIONOFGRAVEGOODSWERETHESAMEINSIDETHECOFFINWEREJADEANDMIRRORSWHIL

EOUTSIDETHECOFFINWEREALARGENUMBEROFTRIANGULAREGDEDIVINEBEASTMIRRORSIRONWEAPONSAGRICULTURALANDFISHINGTOOLSANDSOONREGARDINGTHISAIICHIROSHIRAIISHIBELIEVEDTHATATTHATTIMEVARIOUSFORCESINKINOLEDBYTHEYAMATAIKINGDOMANDTHESETOINLANDSEAFORMEDANALLIANCEANDOBTAINEDIRONANDVARIOUSCULTURALRELICSFROMKOREABYDEFEATINGFORCESSUCHASNOKUKUANDTHEITOKINGDOMSWHICHCONTROLLEDTHEGENKAISEAONTHEOTHERHANDYOSHIROKONDOALSOMENTIONEDTHATMOSTOFTHEZENPOKOENFUNBUILT ON FLATLAND WERE WELL ORGANIZED AND THE MOUNDS WERE ALSO BUILT LAYER BY LAYER AFTER CONSOLIDATION CONSIDERING THE SLOPE AND ARRANGEMENT OF THE MOUNDS CIRCULAR MOUNDS WERE CONSTRUCTED IN ADDITION BRONZE MIRRORS FROM MAINLAND CHINA AND THE WAJINDEN FROM THE SAME PERIOD INDICATE THAT THERE WAS AN EXCHANGE BETWEEN THE TWO COUNTRIES AT THAT TIME THEREFORE IT IS SPECULATED THAT THE EARLY APPEARANCE OF THE ZENPOKOEN FUN MAY HAVE BEEN INFLUENCED BY CIVIL ENGINEERING TECHNOLOGY FROM MAINLAND CHINA AND THE POLITICAL AND IDEOLOGICAL INFLUENCE IS REFLECTED IN THE JOINT CONSTRUCTION OF THE ZENPOKOEN FUN BY VARIOUS ALLIANCE MEMBERS AS PROOF OF THE ALLIANCE AT THE SAME TIME THERE ARE ALSO TOMBS FIVE FOUR AND THREE OF THE KAMENMUN IN THE FRONT OF THE HOUYUAN TOMB IN EASTERN JAPAN DURING THE SAME PERIOD AS ISHIBUTAI KOFUN AMONG THE TOMB FIVE IS THE EARLIEST KOFUN IN EASTERN JAPAN HIS TORY INDICATING A POSSIBLE CONNECTION BETWEEN THE KAZUS A PROVINCE AND THE YAMATAI KINGDOM AT THAT TIME IN THE FOURTH CENTURY HOKURIKU REGION KANTO REGION TOKAI REGION AND TOHOKU REGION WERE MAINLY COMPOSED OF FRONT AND REAR TOMBS WHICH GRADUALLY BECAME ZENPOKOEN FUN TWO TO THREE GENERATIONS LATER REGARDING THIS YAMATO TAKERU POINTED OUT THAT ALTHOUGH THE KOJIKI AND NIHON SHOKI MENTIONED THAT THE FOUR GENERALS AND JAPANESE TAKEZUN AND OTHERS HAD MERGED EAST JAPAN INTO THE TERRITORY OF THE JAPANESE MONARCHY THROUGH MULTIPLE EXPEDITIONS HE BELIEVED THAT IT WAS ACTUALLY KUNUNOKUNI LOCATED IN THE NOBI PLAIN THAT FORMED AN ALLIANCE IN EAST JAPAN MAINLY CONSISTING OF REAR GRAVES IN THE PAST AND THAT AFTER THE DEATH OF HIMIKO THEY LOST THE BATTLE WITH THE YAMATAI KINGDOM OR NEGOTIATED PEACE UNDER THE LEADERSHIP OF THE YAMATO KINGSHIP IN THE END THE ALLIANCE OF THE DOGSLAVE KINGDOM AND THE ALLIANCE OF THE EVIL HORSE KINGDOM MERGED INTO ONE FORMING THE JAPANESE MONARCHY THEN THE LEADERS OF EAST AND WEST JAPAN WERE DISTINGUISHED BY THE ZENPOKOEN FUN AND THE FRONT AND REAR TOMBS FROM THE END OF THE FOURTH CENTURY TO THE FIFTH CENTURY LARGE ZENPOKOEN FUN RESEMBLING KOFUN IN MOUNT TAIDABEGAN TO APPEAR IN PRESENT DAY GUNMA PREFECTURE LARGE ZENPOKOEN FUN HAVE ALSO BEEN BUILT IN PLACES SUCH AS YAMANASHI PREFECTURE IBARAKI PREFECTURE AND CHIBI PREFECTURE AFTER THE MID FOURTH CENTURY THE DAWA TOMBS ORIGINALLY CENTERED AROUND THE YAMATO KOFUN IN THE SOUTHEAST OF THE NARABASIN AND THE YUBEN KOFUN GRADUALLY SHIFTED TO THE DAISEN KOFUN IN THE NORTH THE LARGE ZENPOKOEN FUN OF DAWA WASTHE FURUICHI KOFUN CLUSTER AND MOZUTO MBSTHE CENTER OF GRAVITY OF THE GREAT KING TOMBS SHIFTED AGAIN IN THE FIFTH CENTURY TO THE ANCIENT CITY TOMB GROUP AND THE HUNDRED TONGUE DBIRD TOMB GROUP IN THE OSAKA PLAIN REGARDING THE REASONS FOR THE MULTIPLE TRANSFERS OF THESE LARGE ZENPOKOEN FUN BASED ON RESEARCH ON THE KUSHIJI AND NIHON SHUJI THE JAPANESE ACADEMIC COMMUNITY HAS PROPOSED VIEWSSUCH AS THE THEORY OF DYNASTY ALTERNATION AND THE THEORY OF QUESTRAN TRIBES CONQUERING DYNASTIES TO EXPLAIN ON THE OTHER HAND HIROSE KAZUO PROPOSED THAT THE STRUCTURES OF THE SAKI KOJISHI ZUKI AND MAKINO ANCIENT TOMB GROUPS WERE SIMILAR AND FROM THE LATE FOURTH CENTURY TO THE SECOND HALF OF THE FIFTH CENTURY THE HEADS OF THE FOUR ANCIENT TOMB GROUPS JOINTLY PLAYED A ROLE IN THE YAMATO KINGSHIP AFTER DEATH THEY DEIFIED AS THE GUARDIAN DEITY OF THE YAMATO ROYAL POWER AND ALSO PROPOSED THAT THE CONSTRUCTION OF THE LARGE ZENPOKOEN FUN HAD A DETERRENT EFFECT ON VARIOUS REGIONS AT THE SAME TIME IN THE SECOND HALF OF THE FIFTH CENTURY NO LARGE SCALE ZENPOKOEN FUN WERE CONSTRUCTED IN AREAS SUCH AS KIBI PROVINCE UPPER MAOE AND HYUGA PROVINCE AND THE SCALE OF THE ZENPOKOEN FUN IN GOKISHI CHIDO GRADUALLY DECREASED ACCORDING TO THE INSCRIPTIONS ON JIJITHE IRON SWORD UNEARTHED FROM THE INAWAMOUNTAIN KOFUN AND THE LARGE SWORD UNEARTHED FROM THE ETAFUNAYAMA KOFUN EMPEROR YURYAKU AT THAT TIME REGARDED HIMSELF AS THE KING OF THE WORLD AND HIS AUTHORITY WAS REFLECTED IN THE FACT THAT ONLY HE CONTINUED TO BUILD LARGE ZENPOKOEN FUN IN THE JAPANESE ARCHIPELAGO IN THE SIXTH CENTURY THE HORIZONTAL CAVE STYLE STONE CHAMBER OF THE KINAI SYSTEM BEGAN TO BE POPULARIZED IN ZENPOKOEN FUN AT THE SAME TIME THEY BEGAN TO APPEAR IN AREAS SUCH AS SETSU PROVINCE OWARI PROVINCE UJIPROVINCE AND EVEN IN KANTO REGION WHICH WERE CONSIDERED AS BASES FOR THE ESTABLISHMENT OF THE EMPEROR KEITAI AMONG THEM UENO HAD A PARTICULARLY LARGE NUMBER OF ZENPOKOEN FUN COMPARED TO ONLY THIRTY NINE IN KOZUKE PROVINCE DURING THE SAME PERIOD UENO ALONE HAD NINETY SEVEN WHILE IN KANTO THERE WERE TWO HUNDRED SIXTEEN AS BOTH KINO AND OIZHANG DECREASED ONLY THE KANTO REGION CONTINUED TO CONSTRUCT LARGE ZENPOKOEN FUN THROUGHOUT THE SIXTH CENTURY INDICATING THAT THE YAMATO MONARCHY AT THAT TIME RELIED HEAVILY ON THE EASTERN KINGDOM IN BOTH ECONOMIC AND MILITARY ASPECTS FROM THE END OF THE SIXTH CENTURY TO THE BEGINNING OF THE SEVENTH CENTURY LARGE SQUARE AND CIRCULAR TOMBS REPLACED THE FRONT AND BECAME THE MAINSTREAM IN THE MIDDLE OF THE SEVENTH CENTURY THEY EVOLVED INTO OCTAGONAL TOMBS REGARDING THE REASON FOR THE CESSATION OF CONSTRUCTION OF ZENPOKOEN FUN AIICHIROSHIRAIISHIBELIEVES THAT IT IS RELATED TO THE REFORMS OF CROWN PRINCESHOTO KU AND SOGANO UMAKO AS WELL AS THE ESTABLISHMENT OF THE NATIONAL MANUFACTURING SYSTEM WHILE THE FINAL ZENPOKOEN FUN ARE BELIEVED TO BE AS A MOUNTAIN KOFUN AND RONGMACHI CLAIM THAT THE KOFUN WERE BUILT IN THE FIRST HALF OF THE SEVENTH CENTURY

Kunci

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	RYIOVKJBXNDQFMPHETGSACWLZU

Referensi

Tabel referensi frekuensi monogram, bigram, dan trigram Bahasa Inggris yang digunakan pada langkah-langkah di atas mengacu pada [3, 4].

2.2 Horizon Chartreuse

Deskripsi Soal

Di lembar catatan riset berikutnya, Marisa menemukan laporan mengenai fenomena aneh yang menimpa langit malam. Seseorang telah mencuri bulan purnama yang asli dan menggantinya dengan purnama palsu yang redup. Laporan yang memuat kronologi hilangnya bagian dari langit tersebut kini tidak dapat dibaca karena seluruh teksnya dienkripsi menggunakan **Vigenère Cipher**.

Manfaatkan metode **Kasiski Examination** untuk mengidentifikasi perulangan pola karakter pada *ciphertext* dalam memperkirakan panjang kuncinya. Setelah panjang kunci diperoleh, terapkan analisis frekuensi bahasa Inggris pada tiap sub-teks untuk merekonstruksi kunci dan memulihkan seluruh isi naskah.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1LyCID57V3ckzlvQ9Wnj-mGg_YMH0P50e/view?usp=sharing
- md5sum: 6b933809bbedb3c5c0f1fd9e2b1076cd
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Berkas *Ciphertext*

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Langkah Penyelesaian

Artefak

Plaintext Hasil Dekripsi

Kunci

Referensi

2.3 Domain of Hopes and Dreams

Deskripsi Soal

Dalam suatu catatan simposium, Marisa menemukan rekaman mengenai keberadaan sebuah dimensi buatan yang terisolasi dari dunia luar. Batas-batas menuju wilayah rahasia ini dikunci menggunakan segel jimat yang dienkripsi dengan **Playfair Cipher** agar tidak dapat diakses secara sembarangan.

Tugas kami adalah memulihkan naskah petunjuk akses tersebut. Susun kembali matriks kunci dengan menganalisis pola kemunculan pasangan huruf (*bigram* maupun *n-gram* lainnya), distribusi karakter, serta tebakan kata-kata umum dalam **Bahasa Inggris**. Setelah matriks kunci berhasil direkonstruksi, dekripsikan seluruh dokumen rahasia tersebut.

Berkas lampiran:

- Attachment: <https://drive.google.com/file/d/10Dljf85XJXS41PSSqrfaqU0QyWTuneuX/view?usp=sharing>
- md5sum: b6f21903cbdb62ca899d6bb2694814b1
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Berkas *Ciphertext*

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 SISPDYHMSABRSUTACMRQMSLPRTYBODLOBESPSVTYENDUACBFIDMNDTUIYBNOYCUCTEMDGVCUYSGPPAMZZLTFQATYBFIDYBIAMSIVBG
 TVBCPESACKATCUDBAPEDSACKATCUDBELFSIDGBSTUCRUDITLNUESVBVTBFGUAPFTUKCUYSGSPSGSTCUTOZMLNBQZBCCVCULALDUBDSOL
 UADIUABFTUSIFSCUFEGMYAELUNESYASFYBMNBTTSDBDIOUDMYGMWTVTUZLPRMICVCYUAAHVIOBFIDINYSUAGYTFDUBMWIDSLHKEYACBDI
 NHUAQUDMIEMUGLQSLQPYFZIZGLYIACTVBFEBYMNVBIOICTFMSLELUIBIDBYMNTGSZELCZBQTIOLVTNUFUPAULYTTQSZOSNUFTDNBLDU
 YIYVYSICSOSQTFIATAGBIZTABRIOAPFTIYDIUEBPOUBELDIMSNGIUEBPOUBGINWAMGSYAQNLSPAFFIDBYMNTGUNUBBZTFMNEQBETTYDI
 HBCVGYUNOSVBFWMZVITBWUNTUBSUOCBRTNWUBBZTFMGYAQNTFMQYMTNSNGIUIZBYSZTYUADVBTFIAGLOLIPULIFIOGLOLIPDNLCBDSFBG
 TVSLFIIATDBUTYFTINMYAQNTTACBDPYBQYTOIATCAEBQIOPMMZCNMZTVTIINYSIOZUIFIOTFTIBFIAFPDSMSNGIUYSMSIVDUPSCXOSPR
 YTPRYTTCUGRUIBGBAIPSCXMMWQIOQINWUBALICFLCXLDUMQIZSSISABANFBESYUOELDIPRYSYBPABVQMIYBELCZBQTIISHZOSNUFTDNBUB
 ACITYINPOSYTATPASDQLYGBFTVUFFBDSQAYSVBWNBWMBGPAHYBRBCLUIBBFEBYMNBEGBVTCBDNIYDIYVSDUYIVTGLICUAHYAYTVTOS
 SFBTBFIDBFTVYYSYBBFUAFUCKVTAHBBICVUTCZQAXLBFTVDUACYSVBYIVWUBTTFMWCUTFEIXIBTYACRDSBYSYBBFUAFEQLFCVBYBT
 PSVTQDUKTVYAIQFLLXQLBRBVTVCVCYVDEUAHVBEQMEVFBVBMFYAABQAYAWTTYCRDSFLBTBTFYSLRIOBFIDQDUKTVYAPRYTYUAFEFLUKBF
 CUMWUNTUQAICBFUAHPSUICLUFTPAZLMRZSUKTSGVVTCRDIHBMUNDGCTEGBOSVBSWSDUAAUUKMSNEYAABICGCCYBYTINBTLAQSSFLAIDBUSG
 SZYAXILOBUTYPRYSNNUWBBOPAHXQMBFYAHKCRDSGCCYYBMFIESQKRQSVTYGSFGBVTRUZAMRFGBOIUAESYADNIKUFFBNIEBSIVBIDYSNI
 IFNYTISLQPPRIFSZYUNBSMYBURFSYUBCYRLBFIOYCNWNUQLCONICPUCTEMZNGEONIGBWSTFIATBAUSMBFBVSZABIOATYTSBTELFTLZ
 PQDYCKTUZLOIZLVTBFIDLRIOAPAMUKTVUABFDYZTAUIMVSAZBTFLZIFGBICQLFTDNYBDSUBIZAUZLMFTVTUUKCUYSGPSASLIGBSXPDTY
 YSKTIZTGLNUMIFAESUUKBFTIZMWIQDACDSQDMVUNBHCYIZFTENNIWISUIKSZOLFTYGTIFTNWBWUZCUFRUFIVBTYACRBFVDUYSALWTVBE
 QMEVFBGBVTIOPAMSIVBGTVTPRITPOUBUABFTYLOBVUOBFIDMSNGIUSZQYTONUGCYBCUEAMKTVDRFTVBFTVTRNIFCUBDTIVTAPLADBT
 ICGBYAFIATBVSZABMSEYDUIMPDDUYTTEYDRNIFURBDDUYTTEYDRNIFELDIVTICLFTVTUUFFBAPXBSPPSIPCUYTIAEACKSPSPSIPCUYTUASD
 GBVXSZTOVTINTFOUACQIIICIOSEGBYAFIATBVSZABIAEQCUYSGPADVTYTONZIEBQDAMUKBFIDIOYBNUUCGBCKBEFLUKIFQLBFBZBTINQYC
 IZBTCBCUQSVFDEBAPWFUZZBQDFTVTRNIFCUBDTIVTAPLADBTICGBYAFIATBVSZABDNLCGYAACUAVGIPPQMSYSBFDYBIAITZLPTSPMN
 BEBUSGSZQYCREBIPSAYGBKIAVGTQZSOLGLOLICADQYBANFCVCTIEBCKBEGBICQLFIDPVZVTDEVBNWNIIRIZIONFAPVMAVAVTINVTQSV
 GVIZTBQDOIQYTAIPFLICPZMUGITQSVFSDSYBICASICFLMTIEATPACISZOLNUNUMYUAVGIPPQMSGLFLNUYGUAIOONDBNTVTIDBTMYT
 IHNURUUKYSYBMWTVBVBGTAPDMEODNLCSDBGSGBSLIFGBTYTCKDYHRQLDIGBSTFTIAPRBQLEDQLYBICASAUDIAPBFTVBGWSTFNFAPZL
 IMMAURYUALQIZTESZDUBKMNNDBGMZEODQSYSPBEFLANFBELDIUPZLSUFITPOUBTRMSIYEIEOCUYAHKSQBFBGTVYBFTDNSFMRYAABMS
 LNNFBEMDBDSPBYOETITNMNGEWBYYBBFUANQNDMIETFMSNEYAABCBPBYLSPSSFINPZMQMQSXSXBYBUSEARUAKUMZIZVBVTPACRBUAFNNWIB
 NQBEQLBISUUKQAXLMRTFBKAPZLACUNIBFWMZUACVDBNWUBICNLKXPAINUMYASRSZEIUCIRSZBRSPABFIFGBAINWAMMLDUMRBUUMYAIZDU
 MSNGIUNUGCGLOLAIRFTVINUMYADBSFIUANOUVTOSYNNIWIILHKNZIAMVUOUFEGPRIFQLBRZIOYNYBFTYBURTICZSEADEASIPYSMWIDLR
 CRDSIOGPIINAMZTIMWDQEOZLZACRDSQAUAVGIPPQMSRSPAPZMAUBFUAFNBANFBEMAURODUNVTRSCVBUTREVTQMIECBUAQETIEIWTBTAR
 ODFISDZIIIPDBACUAHVSZABUMOLCREBSIDSNUYGIZTGUABTNUGTUALZSZABRLBFAZPMVZSTTYSZIASEFLUKIDINIBQLFIPMATDUUMYASLHK
 BFUZZBCCVCUBFUAFNUKMEYBYIAUQBTDCBDPYBLSPSSFINIAQDVGUBPRMTMYEIEOUAHKTVSLCREBBKAUGATYEIEIKSZEIAIEBUMLSPPSS
 INITUKCUYSGPMQUZDUAIEIZIUBUCPMMSLYDUIPFLAUQUSLUGIFGBMVUNNMTVASTADNNWUMDPYBDSFNWUMYALTBUUTYGBFSYBCIDBITYRL
 NWBHDUQSWSSQCUEBSFIULPCUDIAUDBUTIIIVBUMYFSEVHUTUCKATCUDINWUMIPLSDUAEBVTIFBTUALSPSSFINTFMVBUIMBDBGPAHYFTIY
 BTDSCEBELDIRFTVQLRUUKCBCUDSNEGBTCMPOTITNMNTYGBIZFIAPRLPAUMFSIUXTFLALIBVITIFTTSPNSTDIACLRITFBSYUZBEQMYTMS
 NGIUMWADVTAUBFGEACTYTVYGCOSYTFASAICRBFBCSDDSPSGHCYUYSYSTNBEMNEQLIFQMYTMAQAPSDRLMFSRTHKDIACMWIDNWBWNO
 YCAIRFTVOUOYIGBFBYLRUNNWUBDQBFYDSIAGYCUFLHKTIVIFUQTYAAACYSLLAPIYDSYBULIFLSUQUADUZMWICZQASLAIEBTCVMSNGIUDM
 BQTIIDERNZIAEOYAQNLQRMQLYGIDDIUABTCUNUHKEBLPCUDIIRSZYBLRIACRDSXTDUEBLZSZABMWTVBEGVTDNLCDUSNNWNIZSRINYOSSF
 QSWTSDKWGLOLIPIAEOYAQNTFQLFIMWIDQITFFLIRIUBFTIUCDBPROIVSDSNUYGYOTSTFINUMSTBKNUYGIRVTDYHRSZSTTYBFIDYLBKTI
 SFIUAPIVYBBFAQOIVTTAKRGVTICELDSIVSLACOIVTINNFTYADVMTSLGMAVSMAUBSIMNIAAUAPLPNQUAUGSIBUYGISNTFLICGBSXPDYYS
 GHURUAIYYOTSTFLINWUMSTBKNUYGYMTYELBRQLDSRTCBFTIDPLAFSSUIRLNAMHSTILATBSIDIACUAFYLDYAFSBUUFASLXTFLIAMODYMN
 BVQMSOOSYTVTNUFUPAPADYRFGVBTIOPAIRSZBFIDTICBDNIYTVBUUFASIPBFDISLYITQITFLVYOVTDQUAQUIMSFAIILAQSSFLAIDBY

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IDEBZSSTFUAINWUMVTNUFUPAZLIFZIUUAHVTFBGBQYINSQYOTSTFLIUHZSRBFLIYBINFTYOSSFYONIUGSIIYBEGBNIEBIFBFDYSTTYDS
FSIUCGBTSDMTATYDSYCSRBDFTPADSCBDPYBYOTSTFLINWUMSTBKIAATQCVBUQLNPSNUCPRMYAUCBTDSIYBEGBSZUTOIACPSZIIMMWNFTL
MRQLFSBUUFASIPVBVYBIAICRDSCTFYIIVEAESTIDSOLGCIYVTFBSYUUKCUYSGPMDDBDSPTYNEBPRIRLNUGCISDYOTSTFLINWUMLSOLEI
VTNUGCBFSYTUICGBYAFIATBVSZABTFFLCXUFBEFELBFDUSTICQLFRZSPMYAIAMVUOITCKGULDZSEVFBTFNYUBFIBKIAVGBFBESONTDSBF
DYIEITTBZMWITOGBOIMYOLAIVGELFIGBNDEIINUMTIUQAQVTITTAUKUABFTYFWUABGTVYTIHIDBYMNBVTFWLSZGUYAQNLSAPFPBFIAGLOL
APUCIFGBYSSFBFIDPLNQAANUGRNIFUAVGIPPQMSLDYAESDUSIQLYGONPRYTTOUADBMFBDLOBEYQCUYSGPSAICVIBFGYTVSEFLUKTIRTUC
IOAMUKIOBFCIZSTTVYOMZTVVTLUAIFFLITOIYSMKTIPSRFCUIYBTEMMSNGIUYSBFIDBFBGLYIAPTYAABAPECTVLNBTYZTVIAEAYTQDPT
PAYGAUCUBIMYSYIAUDDNAIIMMDBDINUBQDQYATYBCSEGBZLSEQMEVFLFIAPWSINDSDPLACRBFYIOBFVTGVAQITPEBDQSQBFVTCYDYMN
BEQLFIPAUHQSVXGBLKLYIAPTYAICQYTOYIZLQPPSSTOUIQZSKVQYQDAMDNFTPAUABFDYIEITTBZMWITAGBIZTABFIOIODUAIRVUADUFIBF
TVNTTYGADUXNSYTFURBEFLCXUAQYUIYONTASIYUMOLAUNNYFTYQZLMSRFLCXUAHEMYBUNUUKBFUAFYFLUKUAQYICGBNUUCQAICMSNGAP
RLOBSAUKBFTIQSVSYA

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2.4 Trinitarian Rhapsody

Deskripsi Soal

Di antara tumpukan lembar catatan risetnya, Marisa menemukan sebuah naskah analisis yang mengulas entitas penguasa tiga alam, sosok yang skalanya melampaui siapa pun di Gensokyo maupun *Lunar Capital*. Naskah yang menyimpan penjelasan mengenai eksistensi mutlak tersebut tidak dapat dibaca karena teksnya telah dienkripsi menggunakan **Affine Cipher**.

Bantulah Marisa mengungkap pesan tersembunyi tersebut tanpa bermodalkan kunci awal maupun *known-plaintext*. Manfaatkan sifat matematis aritmatika modulo pada *Affine Cipher* beserta analisis frekuensi kemunculan karakter dalam bahasa Inggris untuk melakukan **Ciphertext-Only Attack**.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/16aQDPJHcHN2_EmEj2HHTeWeWChkZa_gZ/view?usp=sharing
- md5sum: 48e4f097521e85d43f5b432c67c6d6f9
- sha256sum: 3f10313362d08eb284c954b071345dc92f8c80169f1925d1be2e38971a6e037f

Berkas *Ciphertext*

```

MLLNBUWDHOUALAWNMFMAYNLUHYMVCZLNCMWLNCWKMCFUZEMWDVMIKMLUMZFQOKDMZCOZOJMVQKUZUFMAYNLUHOJUVUQLHIOZMWLNUBCH
LNLNHOUWJOHNHMGVUWEUHUWWCZYQZMVQKUZULNUKOCHMCMZFUCVUCLNICMMWWMJMPAHLONUHMUGUDLMVQKUZUCZQOZLCZAOWBCHLND
MZYWLNUIHUKMCKZUFUWMLUFUMQNGUUDCZYLNUCHMHKQHOWWUFYMCVZLNCMWJUMHCZYLNMMLNUDMCZWOJNUHVMBOAHEOAVFFHCPUVMVQKUZU
KMFHMZLOLNUKOCHMCMZFUCVUCLNICMMZFMZQAZQUFLNMLBIFUWCHUOJXUAWMBOINMFBUUBOZHLOMVQKUZUMZFLNMMLNUCHDHUHOYMLCP
UWNMFBUUZMBOVCWNMFMLMVLNLCWQOZWLHYZMLCOZOJQOAHWUOPUHQMKULNUKOCHMCMZFNLNUICKKUFCLMUVIVULYOLNUCHMHKWMVQKUZUWD
MZYWQUMWUFLMZQUMZFNUHMGVUWEMWBOHZNLUKOCHMCEUHMUYHCUFUFMLLNCMWZFLMOGMEILNUEOKMZVIDMHLWOJYMCVZLNCMWWCZQUB
UCZYBALMKOHLMVWNUUMFFUQUCPUFLNUYOFWLNUILAHZUFNUHCZLOMFUQUCLJAVEUMWUVKMGZYNHVCPUZQHMZZCUWMZFYPUNUHMYYHOL
UWSAUEMIOJKMLCZYNUCWOAZLFLNHOAYNLNUMHWMZFYPUBWCHLNBIBHCZYCYZJOHLNNUHIOAZYLNHOAYNLNUHMLNUQMLUJUVLWO
HHIJOHLNCLWLMZWJOHKMLCOZOJNUHMDUMHMZQUMZFMDDOCZLUFNUHWMWQHUFUWHPMZLOJNUHWUVJCNMPUNUMHFLNMMLNUDOVUQMLEMWOZ
QUMNAKMZBUCZYCLNMWVWUOHUQNUFKINUMHCZYLNMMLYVUEMWNNUHMKULNUZLNMLWNUEMWFMVUHCZWDUVVWMZFMWOHQHUWNLMLWNU

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2.5 Silent Goddess

Deskripsi Soal

Di lembar terakhir catatan risetnya, Marisa menemukan salinan artikel investigasi mengenai sosok dewi misterius bersayap tunggal yang terpergok menyusup ke fasilitas di kedalaman Hutan Bambu. Potongan artikel berita tersebut kini tidak dapat dibaca karena telah dienkripsi menggunakan **Hill Cipher**.

Sang jurnalis yang menyusunnya enggan memberikan matriks kunci secara langsung. Ia hanya membocorkan satu petunjuk bahwa naskah investigasi tersebut diawali dengan kata “**EXPLOSIVE SCOOP**”.

Bantulah Marisa merekonstruksi matriks kunci menggunakan petunjuk awal *plaintext* tersebut melalui **Known Plaintext Attack** (KPA). Hitung invers modulo dari matriks kunci yang ditemukan, lalu gunakan hasilnya untuk mendekripsi sisa *ciphertext* hingga pesan lengkap artikel terungkap.

Berkas lampiran:

- Attachment: <https://drive.google.com/file/d/1aS17ZuSGAWfpcus4F6HEP3wPVXYIHE7n/view?usp=sharing>
- md5sum: 84e86bf1dafb89ec420c114abd60498c
- sha256sum: 6113d33c379b4727e120d3146cfc39ab3004842c0252848dc56c621ba7a5d25f

Berkas *Ciphertext*

```
VNUXGMMXSASCGLRPXOUAIOEKONMNANINLBTMTJFACQMNZAUZIRRYXQOECANINLYACCDKPKICRKFGAZXJKBJOPWJCAISEJCVEECKNINL
BTLPIHPRRFJATIIYQXGJULBVWHMBDPSMEEKGDLUVBAMUIWDOTYJKYZKITJKZJBNOBYSQVSSXMWPGFQAWMQBNBDPSMEIMYXJMWOLWGBLOTMM
XTNJQYFNJVRLJHZOLLRCOJHJZADHACNEVTMSPRFJUIZMSPCOGSARZWDDNAIUBMWUKEXLEPIQGZTICEOWWROCRNOBJCDIIBDOHLBVLZMOJ
KSNKKYYQXPFLRLCNDRMXSCXGKUYQFZQFXFRHTWPUJQWDOTYJKYZOKLPWJTXPMXGMTKCRZSAINMRYUZEVLBVWHMGPVEFWVUBOOKARSI
VYHACPWJSQYGJHASHGQGOUJRCFBFLJJRJATANPQCBKGPVSUENOBNDRCNUAEYSIXIXNFVQMNDSEFKKYZWDGPVBIFMKTACPPGJYCJXJMSMHVM
QEGKOYPINBDNADHUVRPEKZQVCKRNIIZOLQSZTHUDJCDNTBODMGPVZDXPJXRHHKQSFYXWKZIHDKTCVQJJIUVFCVASASKKIVIVYJYDGBSLXC
YFYMDUIZVBXCOBGCHHADGWHLBVWHMUEDACMKRNONEVBZSHYFFHMYGHTKTXPLMZZFTUGSGPVBOTZWDBRLXRH
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Bab 3

Problem Set (Bonus)

3.1 Ex Machina

Deskripsi Soal

Di tengah *hollow night*, sebuah transmisi aneh muncul dari salah satu wilayah yang telah lama ditinggalkan. Sinyalnya berulang secara teratur, tetapi tidak seorang pun mengetahui siapa yang mengirimkannya ataupun sudah berapa lama mesin tersebut aktif.

Setelah diteliti, pesan tersebut diketahui telah dienkripsi menggunakan **Enigma Machine M3**. Namun, setiap percobaan untuk mendekripsinya menggunakan konfigurasi Enigma M3 selalu menghasilkan teks yang tidak masuk akal. Entah karena pengaruh *hollow night*, kerusakan pada mekanismenya, atau memang karena mesin tersebut sejak awal dibangun dengan aturan yang berbeda, Enigma ini telah dimodifikasi.

Analisislah perilaku mesin tersebut, **temukan perubahan yang membedakannya dari Enigma M3 standar**, kemudian lakukan **kriptanalisis** untuk merekonstruksi konfigurasi yang digunakan dan memulihkan pesan aslinya sebelum *hollow night* berakhir.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1TcBU6s876b5ah9EVsuX_xHDTc_0LaWsI/view?usp=sharing
- md5sum: 1074d1cfcaf2482f7229c8f7d40ba097
- sha256sum: fd31f4b0b6475fc15926caea6b28179ea2a1228fe9f063cad8b2d5c215d9031b
- Kode mesin: src/31/enigma-machine/

Berkas *Ciphertext*

```
UTJX CENI RYOW WSVU YEHC YRPR BJWZ NXDJ UNVX WXFE
WOUO DQDZ XJTI QKBI USMI AXEI JFZX XQWJ IBXM PCBW
BAFH OAXG BPBD KASZ LQOY ASNA ZPPC UTVF ISQZ HILW
GYCJ AUQV UGXH VQWW OXMX WEBW XSBQ EGZM OSKQ OYWK
MLKZ YPRC MMHW EHEQ DYJU XLVA CJPN SSYH WFPW WKGS
SAVE XVMF ZGCI SPEJ EFHD RXBS LZDX WYYU UWHB WTJG
ZFJM MTUV HHEZ GVKU IRON NNUU VWFD XZGW BUHB KSSG
VBHP KCQN CJCP GILT KZZK FMLW YVNS OXYJ AXQF RNML
NRYI RLIP CYPC WHQI YLUS JDDH HIDZ OWMH HHML CXXG
BJPY HWFI CMGP XGBP LBHZ ESGG XATC ERAY OSFN MKBP
WWKA PXHU UVIG ICOZ PDLW IMYC XZZH YQUV GUPM WYUE
RYPG TMWO GMTC PMXK ZLPS ACWW UFKO SJST OHGW AXOL
JNBX SQUP UIAP OMLU THXE LCGR HVDP HYLW HXQU OJMO
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NCWD WIYK BZXM RPBO SYGI ZZYF AKCQ HVCN OHBT XEGI
GHXE YBZL OHSB DDZF AZPB AQVL AMCI TBFN TBCS ZNQG
FRCD WZRB KJEL MWDC RDAB NQIH DJKQ PYYB GNLN GWWL
CNCZ NEDW FXXR XPTH LVWN ZCXB TNYH PQUG TNFV AXFE
KSGO HNZZ PKHK HYQL SWQL RWXV LVAC WYWN IFBU JEPX
YSNJ IYXB JVNN QMEI FHMY KJTR YBAM HQBF FMWJ ZZVF
WUMS UPVJ BQUQ OJNP QWMM NABE YCCF RJTQ UHGL MMYM
KMIH VDKC EAOJ YCPB CDQV OEIL FLLO MGHG PPPA QOUS
UYIN MKBE HBGW VGVX ALOJ DGRD FBUQ XYJQ AYHV EKZY
SUJG LADI LTQT DXFZ AHLY LZWZ TBRJ BGKZ NRAB ZKXH
YYTB POXJ MXCP HDPR FCUG CSRB TQQY ZTBH ZZKP XEBO
TLZN XLOY ZSSV MQIU NVMP PSOG HBJZ CTPD EGXV AGWQ
MJBZ AMSI DYKS TZGB LPIW RMKO ZRYY OLTO YKKB YZWJ
EDDR RRLY WZRA PKNF IJGF SBJD IOQW BUCF PUEQ MYOZ
DBVQ GYUB CQJX VCSA TQCY XSUL RFNL ODFC KTKD CCIS
AZFE MHPR SAPV CGFQ HCAB JVBK CSWW CERU UOJX FERO
FGYS PIRG MDTP YWWV BOIK LOBZ QQON QCSN YZPM EEZP
NAJC TDLK VMAL VMRT OIHJ YKGL AYGM YCJE RKZD BBKE
QFNC IIEG JXMQ BMAS KJVG GITP HUTX YTMR YGIA YTHP
DRAF OZEG IHJV SARO HTPJ IVRN XGTJ EXJI MWBK CYCH
NOPN SVMM FRZH RAET IYKA GAKC HFVX BUTT JGGK HKWW
VKBY POVG IYIC KOZA ZDPK VXXE TZZR AJMQ RAVC FJIC
QNNR KJLG ZFMW QOQP KGFT DULW DWWP LHKB UIUP QDCE
CGOG IAIC UCWW GESB YSDW FAFX DYAF QGVZ JBVV VRAF
NAYL JQTN XDYU KANE WKZJ FVAI MSGZ PMKV SRAS PRPD
UVXV NXJI VNAY PVQZ WBPQ FXRG AOGP JXII CPFM ZKWP
IHQY KHOK IHGQ KTQH QTMD MWLG YUUG BOVY QBHG VKXF
LDUG LSWX JAQY SAVN PYFR CCLW OPPB OOMV IHQT SCKN
JTSL ZQLC ISKC LPUI NKJF CQZZ NEDW WXYI SBCO KVPY
RHLV CHMA BOCX GZXT TCHA FJGC GFGU LEGB GESE LXUQ
LZZO BIIP NLYI HDEC OSUH JJZX EPZP PCWP XAVT VNWG
AJQU HXCC ATXZ FMKD NFXC CCSX TNON RNRQ KJQB SODI
HOZQ HCUQ WQJQ YCYD BXHK NSQK RUOM NOCE IXRL PQHZ
NKAJ TAGS ZFSI VQHG ULJV PZPJ KEBX ZXXT UCWX MXBR
MEPQ APOG COGY PJVO MSVW CWOK JYKV MNBR DZTO GATI
XXPN ZHLD OFLX LHWY SFJF IFCZ PDYG BHFQ YCRB GCHA
JAGX PNTK RFOL RWRD ZHPD NOGG JJTO PXPQ RWPU JTQY
YUIE GQJD YCYC MLHM FZLS HNFV QCUY FBCE XXGO FJPT
CQPS YHOD AVPR ALGP JPKB FFDH KLGS ZGXS KYUX BWSX
YPSY MYFY NWYO BNJY FDLR UPNV QRMU YWET RWEF PMCS
SKTK KNVO FPWB QBXA ENXM KMOZ SCPJ SDYX JFCV XPRJ
LDFQ DNCV VMXH WFLH CYIJ XHFC CYAD QRQO AGKT OXGL
VGJC ZFQE MTNZ OBZT AMMC RZMP SFEP YNZO LOIA QSPE
QRVQ LTEM DPZS RJBP YJEV WROV HRFO WJPS BENC AUJH
EDWS GFGJ RNYL BBKG KJXW VLXI EKQL LTUB PSNG XJRR
JZXP PLYP QPVS AXEM QSDI MFOF RMKC PMIU BGCI KMHU
VZRL QRAJ AEUP JQMY SUTY SYCQ FWYQ VDXA PXJV KQBS
HYDU QEUR SZUA FWQF OQDT DUFQ VOKW RWAC BFIF XZNM
VPWU ADXS LNCV LBAK OJNZ JMFO XQWL TZDT SGRB OTKG
DVMA XSQB BDNJ PPGB GIUN JGSV YFER AVPS WDYL XVVA
ZLQB GIBL TGJH DZNH OFKM DXMP VYQG GHIC IZXJ FDBA
XJUL WOZZ XUGC TTHW YGKI NSXI ZREE YYWR IKCS ONQE
NJTZ FZWC ATSM MQLA VLNN BCCS ZBKK FEHZ GTYE ZYVC
XNYB RWXG RHQT RDWK FHJP THQA YPEG ZVQW ZXNX WMRO
YZMU HDDC BAWN MATS XAWM GHDJ TSEJ BOAE EVWU YYMD
YJQU NFEC QLWD WTKG CBTN VPLA DQDY HAYB UPLJ ZHTB
XMLB MTUP AMWM IONR MJRB XOYL PYZU JTRY KATU UAOZ
MLBV JZLL BDYV MXYR OQGX GRUK STVT MLSX OLZJ IEJT
NLRV HNUV PIKX VOBQ EXMU BHZK FHNC UNCN WVEY BHRU
XXSW IXFC DAUC ZYLR IECI ZVDW CTUN FVDE RJZP TRQY
EHMF MDDS DJTE BVIO ZFVO YPXM KKXA SYNS BFFW HOLQ
FZVA IFHS PZJB YAYH CMMD OTLD GEVL LLXJ SEKD WFAD
HYYG ZIIV ZOOM WANE XHND CDTU CGAV JOKK MTLE YDDW
QEFP NIVT CCUL DENX SCSU BPUH GIIT TTCN NCQP UBKC
YCXX PWVO TGVS KNBP GDMS DHIG OMQY IVQH IVCG CARZ
HRXX WSRU WWKO EOQC QKQF GRCF KOQD NRUH MNCT BZGP

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JPGH BYXW RMVV QOWW PNCX BFQM BNGV HLXX WFZE WSAY
SMXY VWOZ LNPV HVHL MSAF EZIJ EVHY EWSI SHUJ VKIG
SNZI IHGV BIGX DELG NXTU KEWJ LJRG YLES JURS TGMN
JUBX QSYU BKAQ YMGM BPBI PCMK UYCY YPIS IVDH NNLO
KNKT MJIE QTKQ IXQF INFF ZOFV NBZS UQLP ZNLF DXCI
FPRW MIHU RRNL VNFQ DCEB MNZH AEAU FRND JIIP KIEJ
WPPI ZPON TRZN BNNT DANG VYEJ OGDY SZVG ZEPZ ZOME
JJJT QHZZ POGO NMYJ DIIZ PQEM YTHF QCFW WTDQ EQML
CENZ ZOOR FUXJ FLZT YEAQ YOEK QTBQ JBDC DLFV PTDA
FFFU DYUS MXPV HBGJ EFJZ POMU HMSJ CYVY QOBV XDEW
LWKV GZXU LHPR WTPV BKMV VESX PFNS PUEN VZCU OEKV
CHJR YODV METT PSYW WFMS EEZW YGJK DBNA KHER SSXE
BZYV NLPL FVRV KUBC YOTP TMOT BCXV GGJU YGYH GVBZ
CLAR HADF VAOX HPGH PVEB XVVY BGTL AMPC UCIL UMCX
FPCP GHGM MXYV IXVH HVTI MUPU XDCN GISL LLMI JUEH
RTMV CKFE ULKD NLYP ZZQC VFUI IWZG TGMH LPSI TEJD
UZTP GBVI XEDU QNCL RIAR UNQA LHAF RFLJ JSTJ BXHL
QSQV XQZA KLLK YGKX BTHY CAPR PORQ DAKA NMYW KHJE
EIGX VGKG NXGL FTEA HEFX BNQU GIKS TJJH CNAJ XQQK
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AZOF SYPB QMSK GSUK HINN TNIJ HRWI KYPM VRBV CLQL
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XQML BINO TTYV JYEE LXOW NXXV ZSLO GMLO ZQHJ OTLZ
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PYCN FPYR AJIB SEGV DSGY TZPD CYMH BCLV FXFG DFSU
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CPUS CFOV ZGKV QSNX ZORB SIGH PXQD WYPY IPQY QTHE
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DKME FPDY DLGR UPFN VWUD AYPC UFCP AVIV QMWU XONL
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CDJG THOB DERH RMZJ BKDK PDFF RPQN UQRG KNFC NMPE
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QLXZ XFOP FZVP ZXGJ RQSS BLPS JDZZ PLSW RCPC MCNG
FCJD PQIO GNCA MYHA GOCF BVZE REZY WIDC DRYX GQJW
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MIQX MIPV PWPA YBMN VWSQ WGJX EPHY ICAB MEMB NMVI
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LQHJ BXFD F

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Analisis Modifikasi Mesin

Langkah Penyelesaian

Artefak

Plaintext Hasil Dekripsi

Kunci / Konfigurasi

Hasil *Decrypt* Sesuai Format Sumber Asli

Referensi

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